

## 16.1 Two-way additive models

Two factors are observed together. Factor  $A$  has levels  $i = 1, \dots, a$  and factor  $B$  has levels  $j = 1, \dots, b$ , with  $a, b \geq 2$ . Every one of the  $ab$  combinations, or **cells**, is observed the same number  $m \geq 1$  of times. We write  $y_{ijk}$  for the  $k$ th observation in cell  $(i, j)$ , so that  $n = abm$ . Example (Additive two-factor model, one observation per cell) found the projections of the additive model for  $m = 1$  by a direct argument. This section redoes it with the tools used throughout the chapter, and adds replication, estimable functions, tests, and a test of additivity with one observation per cell.

### 16.1.1 Averaging and centring matrices

Stack the observations with the replicate index  $k$  running fastest, then  $j$ , then  $i$ :

$$\mathbf{y} = (y_{111}, \dots, y_{11m}, y_{121}, \dots, y_{12m}, \dots, y_{ab1}, \dots, y_{abm})^\top.$$

With this order a matrix of the form  $\mathbf{E}_a \otimes \mathbf{E}_b \otimes \mathbf{E}_m$  acts on each index separately, as in Example (The balanced two-way layout). For  $k \geq 1$  write

$$\bar{\mathbf{J}}_k = \frac{1}{k} \mathbf{1}_k \mathbf{1}_k^\top, \quad \mathbf{C}_k = \mathbf{I}_k - \bar{\mathbf{J}}_k.$$

The **averaging matrix**  $\bar{\mathbf{J}}_k$  replaces each entry of a  $k$ -vector by the mean of the entries. The **centring matrix**  $\mathbf{C}_k$  subtracts that mean. With the usual dot notation for means, the following products replace  $y_{ijk}$  by

$$\begin{aligned} (\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{ij\cdot}, & (\mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{i\cdot\cdot}, \\ (\bar{\mathbf{J}}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{\cdot j\cdot}, & (\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{\cdot\cdot\cdot}, \end{aligned}$$

the cell, row, column and grand means. Each operator averages over the indices where it has a  $\bar{\mathbf{J}}$  and leaves the others alone. Every projection in this chapter is a Kronecker product of factors  $\mathbf{I}$ ,  $\bar{\mathbf{J}}$  and  $\mathbf{C}$ , and the following lemma is all the algebra they need.

#### Lemma 16.1.1. Kronecker products of averaging and centring matrices

Let  $l_1, \dots, l_t \geq 1$ .

- (a)  $\bar{\mathbf{J}}_k$  and  $\mathbf{C}_k$  are symmetric and idempotent, with ranks 1 and  $k - 1$ , and  $\bar{\mathbf{J}}_k \mathbf{C}_k = \mathbf{0}$ ,  $\bar{\mathbf{J}}_k + \mathbf{C}_k = \mathbf{I}_k$ .
- (b) If each  $\mathbf{E}_s$  is one of  $\mathbf{I}_{l_s}$ ,  $\bar{\mathbf{J}}_{l_s}$ ,  $\mathbf{C}_{l_s}$ , then  $\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t$  is an orthogonal projection whose rank is the product of the ranks of the  $\mathbf{E}_s$ .
- (c) If each  $\mathbf{E}_s$  and each  $\mathbf{F}_s$  is one of  $\bar{\mathbf{J}}_{l_s}$ ,  $\mathbf{C}_{l_s}$ , and  $\mathbf{E}_s \neq \mathbf{F}_s$  for at least one  $s$ , then  $(\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t)(\mathbf{F}_1 \otimes \dots \otimes \mathbf{F}_t) = \mathbf{0}$ .
- (d) The  $2^t$  products  $\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t$  with each  $\mathbf{E}_s \in \{\bar{\mathbf{J}}_{l_s}, \mathbf{C}_{l_s}\}$  sum to  $\mathbf{I}$ .

*Proof.*

- (a)  $\bar{\mathbf{J}}_k$  is symmetric and  $\bar{\mathbf{J}}_k^2 = k^{-2} \mathbf{1}(\mathbf{1}^\top \mathbf{1}) \mathbf{1}^\top = \bar{\mathbf{J}}_k$ . Its trace is 1. Then  $\mathbf{C}_k = \mathbf{I} - \bar{\mathbf{J}}_k$  is the complementary projection, with trace  $k - 1$ , and  $\bar{\mathbf{J}}_k \mathbf{C}_k = \bar{\mathbf{J}}_k - \bar{\mathbf{J}}_k^2 = \mathbf{0}$ . Ranks equal traces by Proposition 6.3.4. (b) By Proposition 1.10.1(e) a Kronecker product of symmetric idempotent matrices is symmetric and idempotent, and by Proposition 1.10.1(c) its rank is the product of the ranks. (c) By the mixed-product rule the product equals  $\mathbf{E}_1 \mathbf{F}_1 \otimes \dots \otimes \mathbf{E}_t \mathbf{F}_t$ , and the factor with  $\mathbf{E}_s \neq \mathbf{F}_s$  is  $\bar{\mathbf{J}} \mathbf{C} = \mathbf{0}$  or  $\mathbf{C} \bar{\mathbf{J}} = \mathbf{0}$ . (d) Expand

$\mathbf{I} = (\bar{\mathbf{J}}_{l_1} + \mathbf{C}_{l_1}) \otimes \cdots \otimes (\bar{\mathbf{J}}_{l_t} + \mathbf{C}_{l_t})$  by bilinearity of the Kronecker product. ■

The rank  $k - 1$  of  $\mathbf{C}_k$  is the source of every “levels minus one” in the analysis of variance, and part (d) is the source of every analysis of variance table in this chapter. Part (b) allows  $k = 1$ : then  $\bar{\mathbf{J}}_1 = \mathbf{1}$  and  $\mathbf{C}_1 = \mathbf{0}$ .

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)          # averaging: replaces a k-vector by its
    mean

def Cen(k):
    return np.eye(k) - Jbar(k)              # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    ."""
    P = {}
    for S in itertools.product([0, 1], repeat=len(levels)):
        P[S] = kron_all([Cen(l) if s else Jbar(l) for l, s in zip(levels, S)] + [
            Jbar(m)])
    P["error"] = kron_all([np.eye(l) for l in levels] + [Cen(m)])
    return P

def rank(A):
    return int(round(np.trace(A)))           # rank = trace for a projection
```

[Run this code online](#)

## 16.1.2 The additive model

The additive two-way model is

$$E(y_{ijk}) = \mu + \alpha_i + \beta_j, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad (16.1.1)$$

with model matrix  $\mathbf{X} = [\mathbf{1}_n, \mathbf{Z}_A, \mathbf{Z}_B]$ , where  $\mathbf{Z}_A = \mathbf{I}_a \otimes \mathbf{1}_b \otimes \mathbf{1}_m$  and  $\mathbf{Z}_B = \mathbf{1}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m$  are the indicator matrices of the two factors. It says that the difference between two levels of  $A$  is the same in every column and, symmetrically, that the difference between two levels of  $B$  is the same in every row. The model has  $1 + a + b$  parameters and, by Theorem 8.6.1(a), rank  $a + b - 1$ .

Define

$$\begin{aligned}\mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_B &= \bar{\mathbf{J}}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m,\end{aligned}\tag{16.1.2}$$

and  $\mathbf{R} = \mathbf{I} - \mathbf{P}_0 - \mathbf{P}_A - \mathbf{P}_B$ . By the table of means above,  $\mathbf{P}_A \mathbf{y}$  has entry  $\bar{y}_{i..} - \bar{y}_{...}$  in every position of row  $i$ , and  $\mathbf{P}_B \mathbf{y}$  has entry  $\bar{y}_{.j.} - \bar{y}_{...}$  in every position of column  $j$ .

### 16.1.3 Additivity with one observation per cell

Many two-factor data sets have one observation per cell, such as methods each run once on several test problems. By Example (Additive two-factor model, one observation per cell) the residual space of the additive model then has dimension  $(a-1)(b-1)$ , and the residuals are the interaction terms  $y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{...}$ . The residual sum of squares then serves two purposes. It estimates  $\sigma^2$  if the model is additive, and it is the only place where nonadditivity could show. There is no model-free estimate of  $\sigma^2$ , and no test of additivity against a general alternative.

A test is still possible against a *specific* kind of departure. Suppose the cell means are additive on some other scale,  $\mu_{ij} = f(\nu + a_i + b_j)$ , for a smooth increasing function  $f$  and small effects  $a_i, b_j$  with  $\sum a_i = \sum b_j = 0$ . A second-order Taylor expansion about  $\nu$  gives

$$\mu_{ij} \approx f(\nu) + f'(\nu)(a_i + b_j) + \frac{1}{2}f''(\nu)(a_i^2 + b_j^2) + f''(\nu)a_i b_j.$$

The terms in  $a_i^2$  and  $b_j^2$  are additive. Only the product term is not. To first order the row and column effects of  $\mu_{ij}$  are  $\alpha_i \approx f'(\nu)a_i$  and  $\beta_j \approx f'(\nu)b_j$ , so the interaction is approximately

$$\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{...} \approx \theta \alpha_i \beta_j, \quad \theta = \frac{f''(\nu)}{f'(\nu)^2}.\tag{16.1.3}$$

Tukey (1949) proposed to look for interaction of exactly this shape. The row and column effects are estimated by  $r_i = \bar{y}_{i.} - \bar{y}_{...}$  and  $c_j = \bar{y}_{.j} - \bar{y}_{...}$  (with  $m = 1$  we drop the third subscript). The residuals are then regressed on the products  $r_i c_j$ . The table  $(r_i c_j)$  has zero row and column sums, so it lies in the residual space. The regression through the origin of the residual table on it has slope and sum of squares

$$\hat{\theta} = \frac{\sum_{i,j} r_i c_j y_{ij}}{\sum_i r_i^2 \sum_j c_j^2}, \quad \text{SS}_N = \frac{(\sum_{i,j} r_i c_j y_{ij})^2}{\sum_i r_i^2 \sum_j c_j^2}.\tag{16.1.4}$$

The regressor is computed from the same data, so Proposition 14.7.1 does not apply. But a regressor that is a function of the fitted values alone leaves the  $F$  test exact (Theorem 14.7.2), and part (f) of the next theorem shows that Tukey's regressor is of this kind.

### 16.1.4 Estimation and tests

The following theorem collects what the additive model delivers in a balanced layout, with and without replication.

**Theorem 16.1.2. The additive two-way model**

Assume Eq. 16.1.1 with  $m \geq 1$  observations in every cell.

- (a)  $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B$  and  $\mathbf{R}$  are mutually orthogonal projections summing to  $\mathbf{I}$ , with ranks 1,  $a - 1$ ,  $b - 1$  and  $n - a - b + 1$ . The projection onto  $\mathcal{C}(\mathbf{X})$  is  $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$ , and the fitted values are  $\hat{y}_{ijk} = \bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}$ .
- (b)  $\lambda_0\mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$  is estimable iff  $\lambda_0 = \sum_i \lambda_i = \sum_j \kappa_j$ . The BLUE of a contrast  $\sum_i c_i \alpha_i$  is  $\sum_i c_i \bar{y}_{i..}$ , with variance  $\sigma^2 \sum_i c_i^2 / (bm)$ . These are the estimate and variance of the one-way layout in  $A$  that ignores  $B$ . The estimate of a contrast in the  $\alpha_i$  is uncorrelated with the estimate of any contrast in the  $\beta_j$ .
- (c) Under normal errors,  $SS_A = \|\mathbf{P}_A \mathbf{y}\|^2 = bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$ ,  $SS_B = \|\mathbf{P}_B \mathbf{y}\|^2 = am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$  and  $SSE = \|\mathbf{R} \mathbf{y}\|^2$  are independent,  $SSE/\sigma^2 \sim \chi^2(n - a - b + 1)$ , and

$$F_A = \frac{SS_A/(a-1)}{SSE/(n-a-b+1)} \sim F(a-1, n-a-b+1, \gamma_A), \quad \gamma_A = \frac{bm}{\sigma^2} \sum_i (\alpha_i - \bar{\alpha})^2,$$

with  $\bar{\alpha} = a^{-1} \sum_i \alpha_i$ . So  $F_A$  is an exact test of  $\alpha_1 = \cdots = \alpha_a$ . Its numerator is that of the one-way  $F$  test in  $A$  that ignores  $B$  (Theorem 15.3.1); only the error term differs. The same holds for  $B$ , with  $\gamma_B = (am/\sigma^2) \sum_j (\beta_j - \bar{\beta})^2$ .

- (d) If  $m \geq 2$ , then  $\mathbf{R} = \mathbf{P}_{AB} + \mathbf{P}_E$  with  $\mathbf{P}_{AB} = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m$  and  $\mathbf{P}_E = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m$ , of ranks  $(a-1)(b-1)$  and  $ab(m-1)$ . The squared lengths  $\|\mathbf{P}_{AB} \mathbf{y}\|^2$  and  $\|\mathbf{P}_E \mathbf{y}\|^2$  are the lack-of-fit and pure-error sums of squares of the additive model (Definition 14.6.1). If the cell means  $\mu_{ij}$  are arbitrary, the lack-of-fit  $F$  test of additivity has noncentrality

$$\gamma_{AB} = \frac{m}{\sigma^2} \sum_{i,j} (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..})^2.$$

- (e) If  $m = 1$ , then  $\mathbf{R} = \mathbf{C}_a \otimes \mathbf{C}_b$ , and the residual space consists of the  $a \times b$  tables whose rows and columns all sum to zero. There is no pure error.
- (f) (*Tukey's test.*) If  $m = 1$ , the errors are normal and  $(a-1)(b-1) \geq 2$ , then with  $SS_N$  as in Eq. 16.1.4,

$$F_N = \frac{SS_N}{(SSE - SS_N)/((a-1)(b-1)-1)} \sim F(1, (a-1)(b-1)-1).$$

*Proof.*

- (a) By Lemma 16.1.1(c) the three projections  $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B$  are mutually orthogonal, and their ranks are 1,  $a - 1$ ,  $b - 1$  by Lemma 16.1.1(b). Moreover  $\mathbf{P}_0 + \mathbf{P}_A = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$  replaces each observation by its row mean. It is the projection onto  $\mathcal{C}(\mathbf{Z}_A)$ , the vectors that are constant on rows (Example (The one-way layout)). Likewise  $\mathbf{P}_0 + \mathbf{P}_B$  projects onto  $\mathcal{C}(\mathbf{Z}_B)$ . Hence

$$\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{Z}_A) + \mathcal{C}(\mathbf{Z}_B) = (\mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_A)) + (\mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_B)) = \mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_A) \oplus \mathcal{C}(\mathbf{P}_B),$$

a sum of mutually orthogonal subspaces. By Theorem 6.3.7 its projection is  $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$ . Then  $\mathbf{R} = \mathbf{I} - \mathbf{M}$  is the complementary projection, orthogonal to the other three, with rank  $n - (a + b - 1)$ . The fitted values are  $\mathbf{M} \mathbf{y} = (\mathbf{P}_0 + \mathbf{P}_A) \mathbf{y} + (\mathbf{P}_0 + \mathbf{P}_B) \mathbf{y} - \mathbf{P}_0 \mathbf{y}$ .

- (b) The design is complete, hence connected, and the estimability condition is Exercise

(C1) with a single component. For the estimator,  $E(\bar{y}_{i..}) = \mu + \alpha_i + \bar{\beta}$ , so  $\sum_i c_i \bar{y}_{i..}$  is unbiased for  $\sum_i c_i \alpha_i$  when  $\sum_i c_i = 0$ . It equals  $\boldsymbol{\rho}^\top \mathbf{y}$  with  $\boldsymbol{\rho} = (bm)^{-1} \mathbf{c} \otimes \mathbf{1}_b \otimes \mathbf{1}_m$ , and  $\boldsymbol{\rho} = \mathbf{P}_A \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$  because  $\mathbf{c} \perp \mathbf{1}_a$ . So  $(\mathbf{I} - \mathbf{M})\boldsymbol{\rho} = \mathbf{0}$ , and by Eq. 7.2.1 its variance equals that of the BLUE. By Theorem 7.2.1(c) it is the BLUE. Its variance is  $\sigma^2 \|\boldsymbol{\rho}\|^2 = \sigma^2 \sum_i c_i^2 / (bm)$ . The one-way layout in  $A$  has  $bm$  observations per level, and Theorem 8.7.1 gives the same estimator and variance. A contrast in the  $\beta_j$  has coefficient vector  $\boldsymbol{\rho}_B \in \mathcal{C}(\mathbf{P}_B)$ , and  $\text{Cov}(\boldsymbol{\rho}^\top \mathbf{Y}, \boldsymbol{\rho}_B^\top \mathbf{Y}) = \sigma^2 \boldsymbol{\rho}^\top \boldsymbol{\rho}_B = 0$ .

- (c) The spaces  $\mathcal{C}(\mathbf{P}_0), \mathcal{C}(\mathbf{P}_A), \mathcal{C}(\mathbf{P}_B)$  form an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$  (Definition 9.1.1), so Corollary 9.6.2 applies. The mean vector has entries  $\mu + \alpha_i + \beta_j$ . Its row means minus its grand mean are  $\alpha_i - \bar{\alpha}$ , so  $\|\mathbf{P}_A \boldsymbol{\mu}\|^2 = bm \sum_i (\alpha_i - \bar{\alpha})^2$ . This is zero iff all  $\alpha_i$  are equal. The formula for  $\text{SS}_A$  is the same computation applied to  $\mathbf{y}$ .
- (d) By Lemma 16.1.1(d) with  $t = 3$ ,  $\mathbf{I}$  is the sum of the eight products  $\mathbf{E}_a \otimes \mathbf{E}_b \otimes \mathbf{E}_m$ . The four with  $\mathbf{E}_m = \bar{\mathbf{J}}_m$  are  $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B, \mathbf{P}_{AB}$ . The four with  $\mathbf{E}_m = \mathbf{C}_m$  add up to  $\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m = \mathbf{P}_E$ . So  $\mathbf{R} = \mathbf{P}_{AB} + \mathbf{P}_E$ , with ranks from Lemma 16.1.1(b). The matrix  $\mathbf{I} - \mathbf{P}_E = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$  replaces each observation by its cell mean. It is the projection  $\mathbf{M}_Z$  onto the cell-indicator space, the  $\mathbf{Z}$  of Definition 14.6.1. Hence  $\mathbf{P}_E = \mathbf{I} - \mathbf{M}_Z$  and  $\mathbf{P}_{AB} = \mathbf{M}_Z - \mathbf{M}$ . By Theorem 14.6.2 the noncentrality is  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 / \sigma^2$ , where  $\boldsymbol{\theta}$  has entries  $\mu_{ij}$ . Here  $(\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{P}_{AB}\boldsymbol{\theta}$ , because  $\mathbf{P}_E\boldsymbol{\theta} = \mathbf{0}$ . Its entries are the displayed interaction terms, each repeated  $m$  times.
- (e) With  $m = 1$ ,  $\bar{\mathbf{J}}_1 = \mathbf{1}$  and  $\mathbf{C}_1 = \mathbf{0}$ , so  $\mathbf{P}_E = \mathbf{0}$  and  $\mathbf{R} = \mathbf{P}_{AB} = \mathbf{C}_a \otimes \mathbf{C}_b$ . A vector  $\mathbf{v}$ , read as an  $a \times b$  table  $\mathbf{V}$ , satisfies  $(\mathbf{C}_a \otimes \mathbf{C}_b)\mathbf{v} = \mathbf{v}$  iff  $\mathbf{C}_a \mathbf{V} \mathbf{C}_b = \mathbf{V}$ , which holds iff  $\mathbf{V} \mathbf{1}_b = \mathbf{0}$  and  $\mathbf{1}_a^\top \mathbf{V} = \mathbf{0}^\top$ .
- (f) Write  $\hat{y}_{ij} = \bar{y}_{..} + r_i + c_j$ , let  $\mathbf{r} = (r_i)$  and  $\mathbf{c} = (c_j)$ , and put  $\mathbf{U} = \mathbf{r} \otimes \mathbf{c}$ , the table  $(r_i c_j)$ . Expanding,

$$\hat{y}_{ij}^2 = (\bar{y}_{..}^2 + r_i^2 + 2\bar{y}_{..}r_i) + (c_j^2 + 2\bar{y}_{..}c_j) + 2r_i c_j.$$

The first bracket depends on  $i$  alone and the second on  $j$  alone, so both lie in  $\mathcal{C}(\mathbf{X})$ . Since  $\mathbf{1}_a^\top \mathbf{r} = \mathbf{1}_b^\top \mathbf{c} = 0$ ,  $\mathbf{R}\mathbf{U} = \mathbf{C}_a \mathbf{r} \otimes \mathbf{C}_b \mathbf{c} = \mathbf{U}$  by (e). Hence the column  $\mathbf{W} = (\hat{y}_{ij}^2)$ , a function of  $\mathbf{M}\mathbf{Y}$ , satisfies  $(\mathbf{I} - \mathbf{M})\mathbf{W} = 2\mathbf{U}$ . By Theorem 9.2.2 the extra sum of squares from adding  $\mathbf{W}$  to the additive model is  $(\mathbf{U}^\top \mathbf{y})^2 / \mathbf{U}^\top \mathbf{U}$ , and  $\mathbf{U}^\top \mathbf{U} = \sum_i r_i^2 \sum_j c_j^2$ , so it is  $\text{SS}_N$ , and  $\text{SSE} - \text{SS}_N$  is the residual sum of squares of the augmented model. The rank condition of Theorem 14.7.2 holds because  $\mathbf{U} \neq \mathbf{0}$  with probability one:  $\mathbf{U} = \mathbf{0}$  only if  $\mathbf{P}_A \mathbf{Y} = \mathbf{0}$  or  $\mathbf{P}_B \mathbf{Y} = \mathbf{0}$ , and each event has probability zero, because  $\|\mathbf{P}_A \mathbf{Y}\|^2 / \sigma^2$  has a continuous (noncentral  $\chi^2$ ) distribution on  $a - 1 \geq 1$  degrees of freedom, and likewise for  $B$ . So  $\text{rank}[\mathbf{X}, \mathbf{W}] = a + b$ , which is less than  $n = ab$  because  $(a - 1)(b - 1) \geq 2$ . Theorem 14.7.2 with  $q = 1$  and  $n - r - q = (a - 1)(b - 1) - 1$  gives the distribution of  $F_N$ .

■

Part (b) is the practical content of balance: rows are compared exactly as if the columns did not exist. The columns still matter, because they remove their own variation from the error.

The listing builds the five projections of part (d) for  $a = 3$ ,  $b = 4$ ,  $m = 2$  and checks their ranks, orthogonality and sum. The script also checks each identification in the proof against projections computed from indicator columns.

```
a, b, m = 3, 4, 2
P = factorial_projections([a, b], m)
names = {(0, 0): "mean", (1, 0): "A", (0, 1): "B", (1, 1): "AB", "error": "error"}
for key, name in names.items():
    print(f"{name:6s} rank {rank(P[key])}")
```

```

keys = list(names)
worst = max(np.abs(P[s] @ P[t]).max() for s, t in itertools.combinations(keys, 2))
print("largest entry of any product P_s P_t, s != t:", f"{worst:.1e}")
print("sum is the identity:", np.allclose(sum(P[k] for k in keys), np.eye(a * b * m)
))

```

[Run this code online](#)

## 16.1.5 Tukey's test in practice

Tukey's test (Theorem 16.1.2(f)) is exact against every additive model, whatever the values of  $\mu$ ,  $\alpha_i$ ,  $\beta_j$ . It has power against interactions roughly proportional to  $\alpha_i\beta_j$ , the kind produced by measuring on the wrong scale, and little against anything else. The estimated slope says which scale. If  $f$  is the inverse of a power transformation, so that  $y^p$  is additive, then  $f(t) = t^{1/p}$ , and Eq. 16.1.3 gives  $\theta = (1 - p)/f(\nu)$  (Exercise (B3)). Replacing  $f(\nu)$  by the grand mean suggests the power

$$\hat{p} = 1 - \hat{\theta} \bar{y}_{..}, \quad (16.1.5)$$

with  $\hat{p} = 0$  read as the logarithm. The rule is rough, but it turns a vague complaint into a specific suggestion.

### Example. Solvers on benchmark problems

Six solvers were each run once on eight benchmark problems, and the running time was recorded in seconds. The data are synthetic, generated by the script `tukey_runtime.py` so that the logarithm of time is additive in solver and problem, with a little noise:

|    | P1  | P2  | P3   | P4   | P5   | P6    | P7    | P8    |
|----|-----|-----|------|------|------|-------|-------|-------|
| S1 | 1.7 | 3.4 | 5.6  | 14.1 | 28.1 | 50.2  | 82.9  | 249.4 |
| S2 | 2.1 | 5.7 | 9.0  | 14.7 | 36.9 | 76.0  | 116.7 | 292.0 |
| S3 | 1.3 | 3.1 | 5.1  | 10.5 | 18.8 | 37.6  | 82.2  | 148.3 |
| S4 | 3.3 | 9.1 | 12.7 | 19.1 | 52.8 | 110.0 | 183.1 | 371.8 |
| S5 | 2.0 | 4.7 | 7.2  | 13.5 | 31.4 | 65.2  | 111.1 | 289.3 |
| S6 | 1.5 | 3.4 | 5.4  | 10.4 | 26.0 | 43.2  | 106.3 | 172.0 |

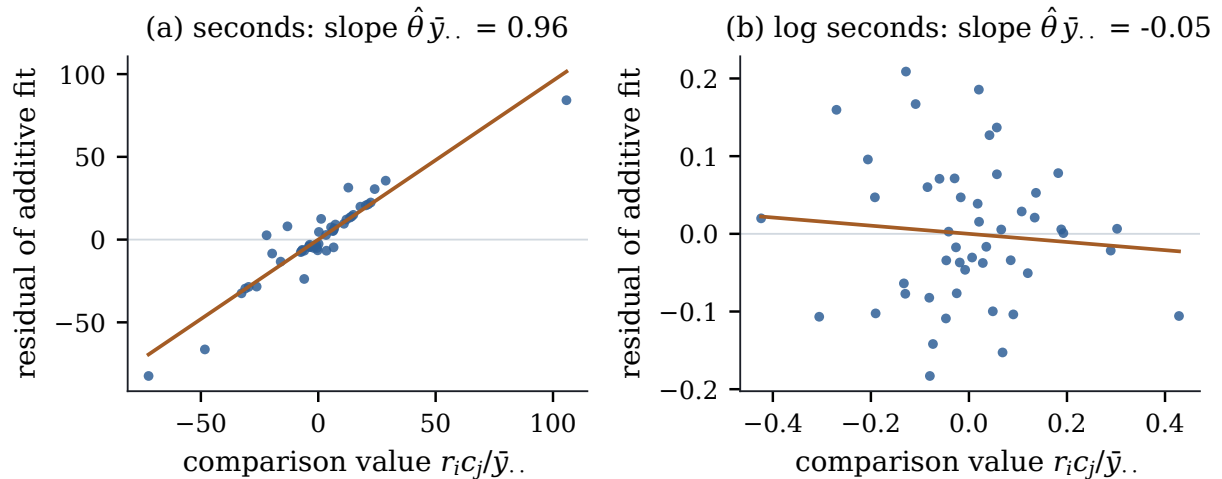


Figure 16.1.1: Tukey's diagnostic plot for the solver data: residuals of the additive fit against the comparison values  $r_{ij}c_j/\bar{y}_{..}$ , with the fitted line of slope  $\hat{\theta}\bar{y}_{..} = 1 - \hat{p}$ . (a) In seconds the residuals lie along a line of slope near one, which points to the logarithm. (b) In log seconds there is no trend.

The additive model in seconds leaves a residual sum of squares of 29396 on 35 degrees of freedom. Tukey's single direction takes  $SS_N = 26126$  of it, a fraction 0.889, and  $F_N = 271.7$  on 1 and 34 degrees of freedom ( $p = 9 \times 10^{-18}$ ). The interaction has exactly the shape of Eq. 16.1.3: fast problems separate the solvers by fractions of a second, slow problems by minutes. The slope is  $\hat{\theta} = 0.01563$  and the grand mean is 61.46, so Eq. 16.1.5 suggests the power 0.04, essentially the logarithm. Figure 16.1.1 shows the diagnostic plot. On the log scale,  $F_N = 0.27$  ( $p = 0.60$ ), and the residuals show no trend against  $r_{ij}c_j/\bar{y}_{..}$ .

The scale changes the conclusion about the solvers. In seconds,  $F_A = 3.88$  ( $p = 0.007$ ), because the nonadditivity inflates the error term. In log seconds,  $F_A = 77.7$ . The residual standard deviation is  $s = 0.105$ , so differences between solvers are measured in proportions of about ten per cent. By Exercise (B1) the solver means of log time are independent with common variance  $\sigma^2/8$  and independent of  $s^2$ . Tukey's method therefore gives simultaneous intervals for all 15 ratios of solver speeds, with multiplier  $q_{0.05}(6, 35) = 4.261$ . Solver 4, the slowest, takes 2.50 times as long as solver 3, the fastest, with simultaneous 95% interval [2.14, 2.94].

```
import numpy as np
from scipy import stats

gen_rng = np.random.default_rng(16_01)
a, b = 6, 8
solver_log = np.array([0.0, 0.25, -0.35, 0.55, 0.1, -0.2]) # log speed
                    factors
problem_log = np.log([2.0, 4.5, 7.0, 12.0, 25.0, 60.0, 110.0, 240.0]) # typical
                    seconds
noise = gen_rng.normal(scale=0.12, size=(a, b))
times = np.round(np.exp(solver_log[:, None] + problem_log[None, :] + noise), 1)

def additive_fit(Y):
    """Row effects, column effects and residuals of the additive fit to an a x b
    table."""
    g = Y.mean()
    r = Y.mean(axis=1) - g # ybar_i. - ybar..
```

```

c = Y.mean(axis=0) - g                                # ybar_.j - ybar..
resid = Y - g - r[:, None] - c[None, :]
return g, r, c, resid

def tukey_test(Y):
    """Tukey's one-degree-of-freedom test for nonadditivity in an a x b table."""
    a, b = Y.shape
    g, r, c, resid = additive_fit(Y)
    u = np.outer(r, c)                                # lies in the interaction space
    ss_n = (u * Y).sum() ** 2 / (u * u).sum()
    sse = (resid ** 2).sum()
    df = (a - 1) * (b - 1) - 1
    F = ss_n / ((sse - ss_n) / df)
    theta = (u * Y).sum() / (u * u).sum()              # coefficient of a_i b_j
    return F, stats.f.sf(F, 1, df), theta, g, sse, ss_n

for label, Y in [("seconds", times), ("log seconds", np.log(times))]:
    F, p, theta, g, sse, ss_n = tukey_test(Y)
    print(f"{label:12s} F = {F:9.2f}  p = {p:.2g}  theta = {theta:.4f}"
          f"   suggested power 1 - theta*ybar = {1 - theta * g:.2f}")

```

[Run this code online](#)

The listing below checks [Theorem 16.1.2\(f\)](#) by simulation. It draws 20000 tables from an additive normal model with  $a = 6$  and  $b = 8$ . The 5% test rejects in a fraction 0.0508 of them, and a Kolmogorov–Smirnov comparison of the simulated statistics with  $F(1, 34)$  gives  $p = 0.98$ .

```

rng = np.random.default_rng(20260919)
reps = 20_000
Fs = np.empty(reps)
for t in range(reps):
    Y = 3.0 + np.arange(a)[:, None] * 0.5 + np.arange(b)[None, :] + rng.normal(size=(a, b))
    Fs[t] = tukey_test(Y)[0]
size = np.mean(Fs > stats.f.ppf(0.95, 1, (a - 1) * (b - 1) - 1))
ks = stats.kstest(Fs, stats.f(1, (a - 1) * (b - 1) - 1).cdf)
print(f"simulated size of the 5% test: {size:.4f}   Kolmogorov-Smirnov p-value: {ks.pvalue:.2f}")

```

[Run this code online](#)

*Warning.* A nonsignificant Tukey test does not establish additivity. It rules out only interaction of the form  $\theta\alpha_i\beta_j$ . A single aberrant cell or row can leave  $F_N$  small and still distort every conclusion, so look at the residual table and an interaction plot ([Section 16.2](#)) as well.

## 16.1.6 Exercises

### 16.1.6.1 A. Check your understanding

#### Exercise. A1

For  $a = b = 2$  and  $m = 1$ , write out the  $4 \times 4$  matrices  $\mathbf{P}_0$ ,  $\mathbf{P}_A$ ,  $\mathbf{P}_B$  and  $\mathbf{R}$  of Eq. 16.1.2. Show that  $\mathbf{R}\mathbf{y} = \frac{1}{4}(y_{11} - y_{12} - y_{21} + y_{22})(1, -1, -1, 1)^\top$ , and explain why Tukey's test is not available here.

### Exercise. A2

In the additive model with  $m$  observations per cell, show that the sum of squares for testing a single contrast  $\sum_i c_i \alpha_i = 0$  is  $(\sum_i c_i \bar{y}_{i..})^2 / (\sum_i c_i^2 / (bm))$ , the same as in the one-way layout in  $A$  (Proposition 15.2.1). What differs between the two  $F$  tests of this contrast, the one here and the one-way test that ignores  $B$ ?

## 16.1.6.2 B. Practice

### Exercise. B1

In the additive model, suppose  $\alpha_1 = \delta$  and  $\alpha_2 = \cdots = \alpha_a = 0$ . Find  $\gamma_A$ . With  $a = 4$ ,  $b = 5$  and  $\delta = \sigma$ , compare the power of the 5% test for  $m = 1$  and  $m = 2$  using Theorem 11.5.1, and explain why doubling  $m$  more than doubles the error degrees of freedom.

*Solution.* Here  $\bar{\alpha} = \delta/a$ , and  $\sum_i (\alpha_i - \bar{\alpha})^2 = \delta^2(1 - 1/a)^2 + (a - 1)\delta^2/a^2 = \delta^2(a - 1)/a$ . So  $\gamma_A = bm\delta^2(a - 1)/(a\sigma^2)$ . With  $a = 4$ ,  $b = 5$ ,  $\delta = \sigma$  this is  $3.75m$ . The error degrees of freedom are  $n - a - b + 1 = 20m - 8$ , which is 12 for  $m = 1$  and 32 for  $m = 2$ . The power is  $\Pr\{F(3, 20m - 8, 3.75m) > F_{0.05}(3, 20m - 8)\}$ , about 0.25 for  $m = 1$  and 0.56 for  $m = 2$  (computed by the script `exercise_checks.py`). The noncentrality doubles, and the error degrees of freedom more than double because the  $a + b - 1$  degrees of freedom spent on the mean structure are a fixed cost. Both effects raise the power.

### Exercise. B2

Show that Tukey's statistic  $F_N$  is unchanged if  $\mathbf{y}$  is replaced by  $\lambda\mathbf{y} + \mathbf{t}$ , where  $\lambda \neq 0$  and  $\mathbf{t}$  is any additive table  $t_{ij} = u_i + v_j$ . Conclude that  $F_N$  has the same distribution under every additive model with the same  $\sigma^2$ .

### Exercise. B3

Suppose  $y^p$  is additive for some  $p \neq 0$ , so that  $\mu_{ij} = f(\nu + a_i + b_j)$  with  $f(t) = t^{1/p}$ . Verify that  $\theta = f''(\nu)/f'(\nu)^2 = (1 - p)/f(\nu)$ , and derive Eq. 16.1.5. What does  $\hat{\theta}\bar{y}_{..} \approx 1$  suggest, and why is  $p = 0$  the right reading?

*Solution.*  $f'(t) = p^{-1}t^{1/p-1}$  and  $f''(t) = p^{-1}(p^{-1} - 1)t^{1/p-2}$ . So

$$\frac{f''(\nu)}{f'(\nu)^2} = \frac{p^{-1}(p^{-1} - 1)\nu^{1/p-2}}{p^{-2}\nu^{2/p-2}} = p\left(\frac{1}{p} - 1\right)\nu^{-1/p} = \frac{1 - p}{f(\nu)}.$$

Solving  $\theta = (1 - p)/f(\nu)$  for  $p$ , and estimating  $\theta$  by  $\hat{\theta}$  and  $f(\nu)$ , the typical size of a cell mean, by  $\bar{y}_{..}$ , gives  $\hat{p} = 1 - \hat{\theta}\bar{y}_{..}$ . A value  $\hat{\theta}\bar{y}_{..} \approx 1$  gives  $\hat{p} \approx 0$ . For  $f(t) = e^t$ , which makes  $\log y$  additive,  $f''/f'^2 = e^{-\nu} = 1/f(\nu)$ , which is the limit  $p \rightarrow 0$  of the power formula. This

matches the convention that  $(y^p - 1)/p \rightarrow \log y$ .

#### Exercise. B4

With  $m = 1$  and normal errors, suppose the cell means are  $\mu_{ij} = \mu + \alpha_i + \beta_j + \theta\alpha_i\beta_j$  with  $\sum_i \alpha_i = \sum_j \beta_j = 0$ , exactly the shape of Eq. 16.1.3. Show that  $r_i = \alpha_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$ , and that, given  $\mathbf{P}_A \mathbf{Y}$  and  $\mathbf{P}_B \mathbf{Y}$ ,

$$E(\hat{\theta} \mid \mathbf{P}_A \mathbf{Y}, \mathbf{P}_B \mathbf{Y}) = \theta \frac{\sum_i r_i \alpha_i}{\sum_i r_i^2} \cdot \frac{\sum_j c_j \beta_j}{\sum_j c_j^2}.$$

Explain why  $\hat{\theta}$  is nearly unbiased when the row and column effects are large compared with  $\sigma$ , and why it tends to be pulled towards zero when they are not. What does this imply for the power rule Eq. 16.1.5?

*Solution.* The interaction terms  $\theta\alpha_i\beta_j$  have zero row and column means, so  $\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot} = \alpha_i$  and  $r_i = \alpha_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$ ; likewise for  $c_j$ . With  $\mathbf{U} = \mathbf{r} \otimes \mathbf{c}$  as in the proof of Theorem 16.1.2(f),  $\hat{\theta} = \mathbf{U}^\top \mathbf{R} \mathbf{Y} / \mathbf{U}^\top \mathbf{U}$ . By Theorem 4.4.4,  $\mathbf{R} \mathbf{Y}$  is independent of  $(\mathbf{P}_A \mathbf{Y}, \mathbf{P}_B \mathbf{Y})$ , whatever the mean, so given those two vectors  $\mathbf{U}$  is fixed and  $E(\mathbf{R} \mathbf{Y}) = \mathbf{R} \boldsymbol{\mu}$ , the table  $(\theta\alpha_i\beta_j)$ . Hence the conditional mean is  $\theta \sum_{i,j} r_i c_j \alpha_i \beta_j / (\sum_i r_i^2 \sum_j c_j^2)$ , which factorizes as displayed. If the effects dominate the noise,  $\mathbf{r} \approx \boldsymbol{\alpha}$  and both factors are near one. If not,  $\sum_i r_i^2$  carries the extra noise  $\sigma^2(a-1)/b$  on average while  $\sum_i r_i \alpha_i$  does not: the attenuation of a regression on a noisy regressor. The power rule then suggests a  $\hat{p}$  too close to one, too weak a transformation.

#### 16.1.6.3 C. Going deeper

#### Exercise. C1

Extend Tukey's test to  $m \geq 2$  observations per cell. Replace  $y_{ij}$  by the cell mean  $\bar{y}_{ij}$  in Eq. 16.1.4 and multiply the result by  $m$ , and compare it with the pure-error mean square. Show that under the additive model the resulting statistic has the  $F(1, ab(m-1))$  distribution. Show also that  $SS_{AB} - SS_N$  gives a test of the remaining nonadditivity.